



ENT-164 · Intro to Making



Derby Entrepreneurship
Center at Tufts

CLASS 6 · WEEK 6 · FALL 2026

Coding with an AI Agent

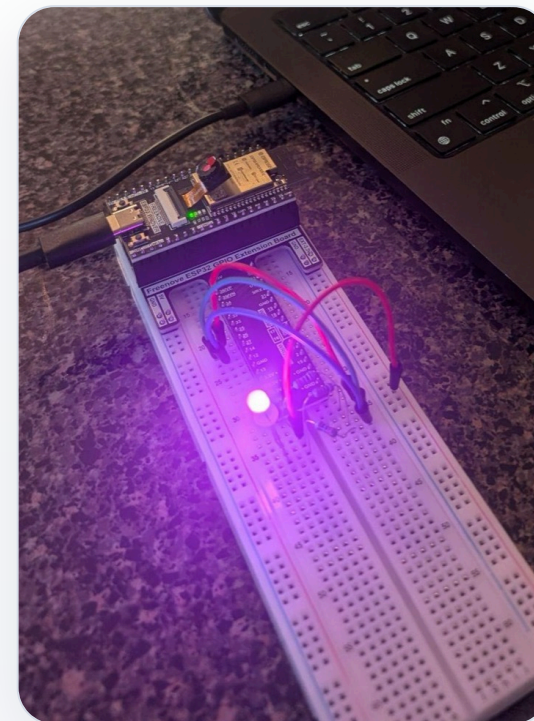
Your ESP32 can serve a web page your phone opens. No app required.
Today we get it on WiFi, register it with Tufts, and build a smart light you
control from the browser.

Thu, Oct 15

JCC 301 / Nolop

No programming required

Bring your ESP32



Before we go **wireless**

You should have all three from last class. Tell us now if something is missing.

01

Electronics kit

You've received your electronics kit. The parts you'll wire up today.

02

opencode + MicroPython

opencode is set up and MicroPython is deployed to your ESP32. It runs your code on the board.

03

AI-assisted electronics

You can already build basic electronics projects with the help of AI.

From your phone **to the internet**

Connect, reach out, and build. The three things we do in class today.

01

Connect to your microcontroller

Reach your device from a phone and from a web browser. No native app needed.

02

Connect to external services

Give your device internet access, higher data rates, and cloud services.

03

Build a Smart Light

Put it all together: an RGB light you control from your phone's browser.

Two ways to connect

Both work. Only one of them skips app development entirely.

Bluetooth

Short-range, device to device

Low-power, short-range communication. Like connecting to a phone app. Requires a native mobile app for user interaction.

This is hard. You'd have to build and ship an app.

WiFi

Internet-connected, browser-first

Internet access, higher data rates, and integration with cloud services. Users interact through a simple web browser interface. No app development needed.

Today's approach. The browser is your user interface.

Client, server, or **hybrid**

Three pieces: your device, the services it reaches, and the browser you control it from.



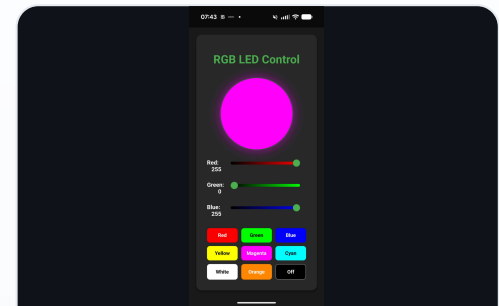
The device

An ESP32 WROVER running MicroPython. The thing that gets online.



External services

Over WiFi, your device can reach APIs and cloud services.



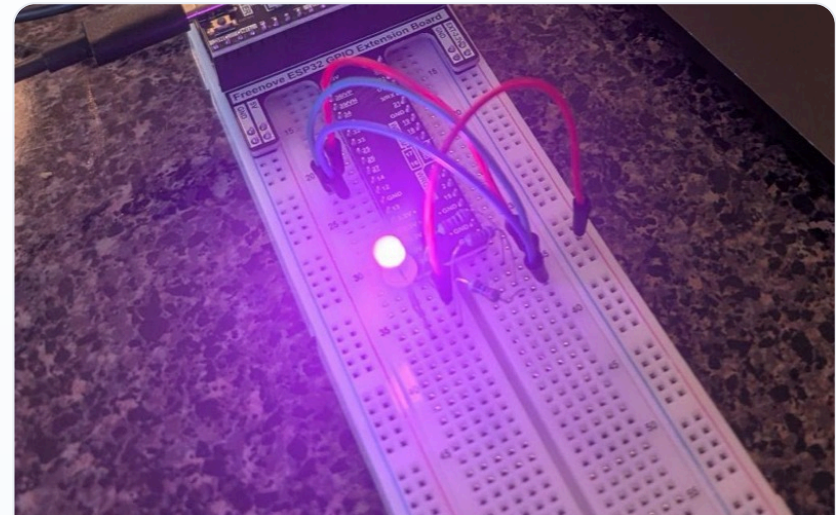
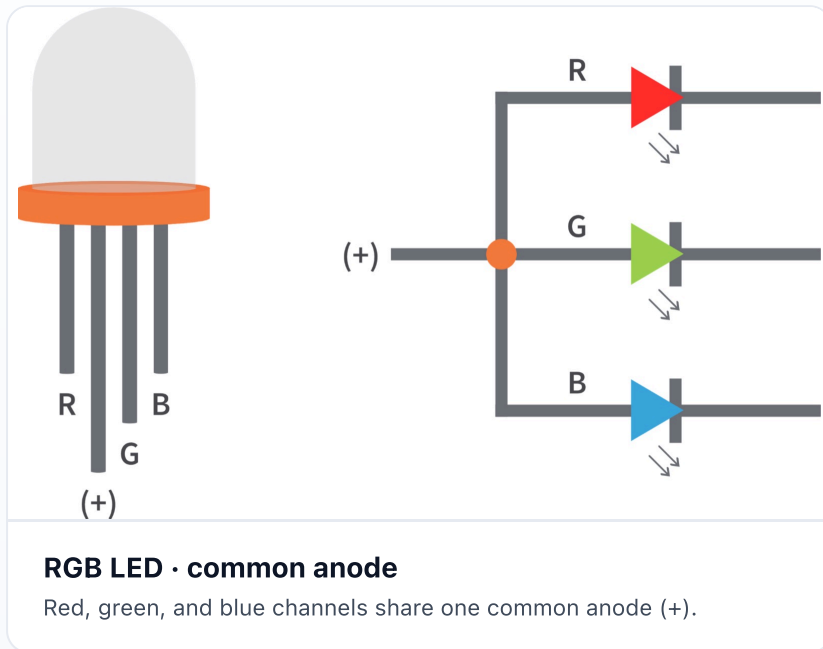
The browser UI

No app to install. The device serves a page your phone opens.

Client. Your device reaches out to a service · **Server,** your device serves a page · **Hybrid,** the Smart Light is both.

Wiring the Smart Light

One RGB LED. Three colors in a single package, sharing a common anode (+).



On the breadboard

Each channel gets its own pin. The page mixes them into any color.

Ask for it in plain English

No programming required. Describe the project and let AI write the MicroPython.

THE PROMPT

"I want to use an ESP32 WROVER with MicroPython to measure the distance with a HC-SR04 sensor and show it on my phone using the Tufts_Wireless open wifi hotspot. Print my ESP32 MAC address on startup."

WHAT THE AI NEEDS FROM YOU

THE BOARD & LANGUAGE

ESP32 WROVER, MicroPython

THE COMPONENT

HC-SR04 distance sensor

THE GOAL

Measure distance, show it on your phone

THE CONNECTION

The Tufts_Wireless open WiFi hotspot

The MAC line is extra: it prints the address you'll register with Tufts IT. Next slide.

Register your ESP32 with Tufts IT

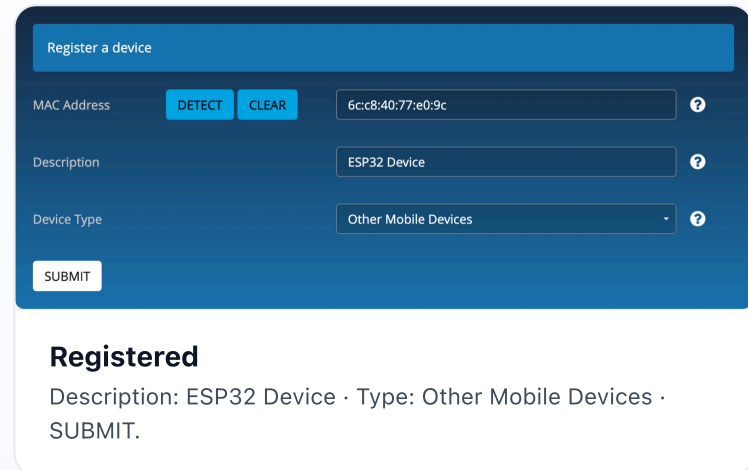
Tufts_Wireless needs your device's MAC address before it will let it join.

01 Run the code with opencode and copy your device's MAC address from what it prints. It looks like `6c:c8:40:77:e0:9c`, but with your own numbers and letters.

02 The rest of the code will fail at this point. You can't reach WiFi yet. That's expected.

03 Go to <https://device-registration.it.tufts.edu>.

04 Clear the default MAC address, then copy your ESP32's MAC from opencode into the field.



The screenshot shows a web form titled "Register a device" with a dark blue header. Below the header, there are three input fields: "MAC Address" with "DETECT" and "CLEAR" buttons, "Description" with a text input, and "Device Type" with a dropdown menu. Each field has a help icon (question mark). The "MAC Address" field contains the text "6cc8:40:77:e0:9c". The "Description" field contains "ESP32 Device". The "Device Type" dropdown is set to "Other Mobile Devices". At the bottom of the form is a "SUBMIT" button. Below the form, the text "Registered" is displayed in bold, followed by "Description: ESP32 Device · Type: Other Mobile Devices · SUBMIT."

Use the IP address, **not the MAC**

opencode prints two addresses. Only one of them belongs in your browser.

Use this

192.168.6.134

Type the IP address opencode prints into your phone's browser, e.g. `http://192.168.6.134`. That opens your device's page.

Not this

6c:c8:40:77:e0:9c

The MAC address was only needed for Tufts IT registration. It will not open a web page.

ATTENTION It may take a few minutes before your device registration is active. If the page doesn't load, wait and refresh.

What happened?

Every tap on your phone becomes a request the ESP32 answers.

A web server starts

1

Your ESP32 runs a web server that renders a web page. No separate computer needed.

The page gets an address

2

The page is accessible at the device's IP address, assigned automatically when it joins Tufts_Wireless.

Your phone renders it

3

Your phone's browser renders the HTML that the ESP32's web server generates.

Requests round-trip

4

Interacting with the page sends a request to the ESP32 (e.g. `http://10.243.94.14/`). It reads the request, calls the sensor, and passes the data back.

Other things to try

Once your device is online, the browser becomes a remote control for anything you build.

01

Visualize sensor data

A dashboard with temperature history charts.

02

Control devices from your phone

A virtual joystick to control motors, or a virtual switch to turn something on or off.

03

Manage settings on the phone

Save them to the device: set a thermostat temperature, or set an alarm clock.

04

Device state visualization

See the status of an LED (on/off) or the current servo angle.

MILESTONE REFLECTION · 2 OF 5

After connectivity

Complete your milestone reflection for this phase of the course before you leave today.

go.tufts.edu/reflect

BEFORE YOU GO

Questions?

Then get your device online. Next week: Final Robot Assembly and Testing. Integrate everything you've learned into a working robot.

Individual assignment: electronics project. Present a working circuit.

WHERE TO FIND US

Office hours

`go.tufts.edu/meet_tvdv`

Email

`thomas.van_de_velde@tufts.edu`

Course materials

`tuftsmaker.github.io/ENT-164`